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Improved structure-adaptive anisotropic filter

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Abstract

Most adaptive smoothing approaches damage image fidelity. In this paper we propose to improve the structure-adaptive anisotropic filtering approach using an elliptical kernel, a non-linear filtering function, and a more robust-to-noise technique for oriented pattern direction estimation. The proposed filter outperforms some others known filters and is more robust to noisy images.
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Keywords: Structure-adaptive anisotropic filter; Image smoothing; Orientation estimation

1. Introduction

Noise filtering of images is basically a smoothing process, and it is a subject that has been addressed for many years. The idea of adaptive smoothing is being investigated a long time and many different approaches have been proposed over the years. Mastin (1985) presents a detailed overview and evaluation of some of earlier methods. One of the earlier studies in this field was performed by Zucker et al. (1977). They have suggested to apply at each image point a weighted mask with coefficients, which are based on the differences between the value at the center point and those of its neighbors. Similar approaches are proposed in some other studies (e.g. Wang et al., 1981; Davis and Rosenfeld, 1978). The major drawback of these iterative smoothing methods is that their convergence properties are difficult to establish. The approach, proposed by Blake and Zisserman (1987), uses weak continuity constraints to allow discontinuities in a piecewise continuous reconstruction of a noisy signal. The convergence behaviour was studied, but the approach suffers from a formidable computational cost. By casting the problem in terms of a heat equation in anisotropic medium, Perona and Malik (1990) presented an anisotropic diffusion scheme. The principal aim of their method is to enhance edges and facilitate region segmentation, however, when small iteration steps are used it can also be considered as a useful tool for image noise filtering (Gerig et al., 1992). Almansa and Lindeberg (2000) and Weickert (2001) have used diffusion techniques, which are based on a multi-scale analysis called scale-space representation, and applied an iterative process for local features estimation. Diffusion methods have two major drawbacks (Yang et al., 1996): they slow down the filtering process in images with considerable amount of noise, and they tend to distort sloping edges. Steerable filters are a class of filters, in which a filter of arbitrary orientation is synthesized as a linear combination of a set of "basis filters" (Freeman and Adelson, 1991). Steerable filters are used in many vision and image processing tasks, and the image enhancement is one of them. Steerable-scalable kernels roughly shaped like Gabor functions have the advantage that they can be specified and computed easily (Perona, 1992). However, current steerability approaches suffer from the consequences of the uncertainty principle, and in order to achieve a high orientational resolution a huge number of basis filters must be used (Yu

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et al., 2001). Another kind of structure-adaptive anisotropic filtering technique has been proposed by Yang et al. (1996). Instead of using local gradients as a means of controlling the anisotropism of filters, it uses both a local intensity orientation and an anisotropic measure to control the shape of the filter. Although the filters proposed by Yang and Almansa are both structure-adaptive anisotropic filters, they still significantly differ.

We propose to improve the structure-adaptive anisotropic filter (Yang et al., 1996) in the space domain. We suggest changing the filter's kernel from a circle to an ellipse with the form, size and direction depending on image local anisotropic features. The essential idea of this paper is to apply a median filter for pixels bounded by an anisotropic elliptical kernel. We propose to use a non-linear filter kernel function rather than a linear one, which produces less blurring during image filtering. The non-linear function is implemented in the form of the median filter. Moreover, instead of using Yang et al.'s (1996) derivatives-based method for estimation the oriented pattern direction we have used Donahue and Rokhlin's (1993) method, which is more robust to noise. It uses a gradient type local operator and least-squares minimization to control the noise. Comparison results on images contaminated by a Gaussian noise and 'Salt and Pepper' noise show superior performance of the improved structure-adaptive anisotropic filter over the Yang's structure-adaptive anisotropic filter and over the conventional median one. The paper is organized as follows. Section 2 reviews the structure-adaptive anisotropic filter proposed by Yang et al. (1996). In Section 3 we present some improvements in the space domain to the structure-adaptive anisotropic filter. Section 4 presents the comparison study of applying our improved structure-adaptive anisotropic filter, the Yang's structure-adaptive filter and the median filter on images contaminated by a Gaussian noise and 'Salt and Pepper' noise. Finally, in Section 5 some conclusions are drawn.

2. The structure-adaptive anisotropic filter

The structure-adaptive anisotropic filter, which has been proposed by Yang et al. (1996), uses a local intensity orientation and an anisotropic measure of level contours to control the shape and extent of the filter kernel. The filter kernel applied at each point x_0 is defined as follows (Yang et al., 1996):

$$k(x_0,x) = \rho(x-x_0) \times \exp\left\{-\left[\frac{((x-x_0)\cdot n)^2}{\sigma_1^2(x_0)} + \frac{((x-x_0)\cdot n_\perp)^2}{\sigma_2^2(x_0)}\right]\right\} \tag{1}$$

where n and $n_\perp$ are mutually normal unit vectors, and n is parallel with the local oriented pattern direction. The shape of the kernel is controlled through $\sigma_1^2(x_0)$ and $\sigma_2^2(x_0)$, ρ satisfy the condition $\rho(x) = 1$ when $|x| < r$, and r is the maximum support radius.

Direction estimation of an oriented pattern is based on the fact that the power spectrum of such a pattern lies along a line through the origin in the Fourier domain, and the direction of the line is perpendicular to the dominant spatial orientation of the pattern. The evaluation of the Fourier transform, however, is not necessary for actual calculations. Through a simple relationship between the local orientation direction and the matrix eigenvectors, the estimation of oriented pattern direction $\theta(x)$ (the direction of vector n in (1)) can be done (Yang et al., 1996) as follows:

$$\theta(x) = \frac{1}{2}\tan^{-1}\left\{\frac{\int\int_\Omega 2\cdot\left(\frac{\partial f}{\partial x_1}\right)\left(\frac{\partial f}{\partial x_2}\right)\mathrm{d}x_1\,\mathrm{d}x_2}{\int\int_\Omega \left(\frac{\partial f}{\partial x_1}\right)^2 - \left(\frac{\partial f}{\partial x_2}\right)^2 \mathrm{d}x_1\,\mathrm{d}x_2}\right\} + \frac{\pi}{2} \tag{2}$$

where Ω is a local neighborhood of $x = (x_1, x_2)$.

The space constants $\sigma_1^2(x_0)$ and $\sigma_2^2(x_0)$ are controlled through the corner detector $c(x_0)$ and by the measurement of anisotropism $g(x_0)$ as follows (Yang et al., 1996):

$$\sigma_1(x_0) = \frac{r}{1 + c(x_0)/\beta} \tag{3}$$

$$\sigma_2(x_0) = (1 - g(x_0))\frac{r}{1 + c(x_0)/\beta} \tag{4}$$

where β is a normalization factor that controls how faithfully the corners and junctions must be preserved during the filtering process.

The anisotropic measure gives an indication of how strong a pattern is oriented and is defined as follows (Yang et al., 1996):

$$g(x) = \frac{\left\{\int\int_\Omega \left(\frac{\partial f}{\partial x_1}\right)^2 - \left(\frac{\partial f}{\partial x_2}\right)^2 \mathrm{d}x_1\,\mathrm{d}x_2\right\}^2 + \left\{\int\int_\Omega 2\cdot\left(\frac{\partial f}{\partial x_1}\right)\left(\frac{\partial f}{\partial x_2}\right)\mathrm{d}x_1\,\mathrm{d}x_2\right\}^2}{\left\{\int\int_\Omega \left(\frac{\partial f}{\partial x_1}\right)^2 + \left(\frac{\partial f}{\partial x_2}\right)^2 \mathrm{d}x_1\,\mathrm{d}x_2\right\}^2} \tag{5}$$

which can be calculated directly from the original data $f(x_1, x_2)$ and its partial derivatives.

The anisotropic measure can also provide a convenient way of finding corner and junction points within a given image. Yang et al. (1996) suggests using both the measure of anisotropism and a gradient strength for an estimation of corner strength in a following way:

$$c(x) = (1 - g(x))|\nabla f(x)|^2 \tag{6}$$

3. Improved structure-adaptive anisotropic filter

In this section we propose some improvements to the structure-adaptive anisotropic filter (Yang et al., 1996) in the space domain. As previously mentioned the essential idea of this paper is to apply the median filter for pixels bounded by an anisotropic elliptical kernel.

The structure-adaptive anisotropic filter suffers from the following problems: corner strength measure $c(x)$ is highly influenced by noise. This results in a wrong estimation of space constants $\sigma_1^2(x_0)$ and $\sigma_2^2(x_0)$, which control the shape of the filter kernel. Derivatives-based approach for oriented pattern direction estimation fails to produce correct estimates for noisy images. The normalization factor β controls how faithfully the corners and junctions can be preserved during the filtering process. Therefore, it is a critical factor and the choice of β significantly affects the filter performance. Despite the fact that the structure-adaptive filter is directional and adjusts the shape of the kernel according to image anisotropic local features, the filter causes unnecessary blurring in processed image due the linearity of its filtering function. The structure-adaptive anisotropic filter operates on a pixels neighborhood of a constant size and moreover, the size does not depend on local features of input image. We suggest a solution to the above-mentioned problems and propose the improved structure-adaptive anisotropic filter, which combines non-linear filtering function, a more robust-to-noise technique for oriented pattern direction estimation and elliptical kernel with its form, size and direction depending on image local anisotropic features.

Instead of using a constant kernel the size must be changed to embody image local anisotropic features, namely, the anisotropic measure $g(x)$ and the corner strength $c(x)$. This can be achieved by defining an elliptical kernel (Fig. 1) with its principal axis and direction changing according to image local anisotropic features.

The elliptical kernel's main axis $a(x_0)$ must be minimal in regions where there is a high number of corners (edges, corners and etc.) and maximal in regions with no corners (smooth places). The transition of $a(x_0)$ from smooth regions to regions that include corners should be performed in an exponential manner in order to prevent smoothing corners. The kernel's shape must be circular in regions with low values of anisotropic measure ($g \to 0$). In regions with high anisotropic measure ($g \to 1$) the shape obtains a highly oriented elliptical form with its main axis runs in parallel to direction of local oriented pattern as can be seen in Fig. 2. To meet these requirements we propose to define the principal axes of the elliptical kernel as follows:

$$a(x_0) = r \cdot \exp\left(\frac{-c(x_0)}{\beta}\right) \tag{7}$$

$$b(x_0) = a(x_0) \cdot (1 - g(x_0) + \varepsilon) \tag{8}$$

where r is the maximal support radius, ε is the minimal kernel width and β is a normalization factor that will be defined later.

Comparing the behaviour of Yang's space parameter $\sigma_1(x_0)$ (Eq. (3)) with the new proposed $a(x_0)$ parameter (Eq. (7)) highlights the improvement of the new filter. Fig. 3 demonstrates the behaviour of controlling the main axis of the filter kernel for both Yang's and the proposed filter. This figure shows that adopting the proposed filter results with better behaviour relative to the corner strength $c(x)$. For high values of corner strength the elliptical kernel results with smaller values for the main axis $a(x_0)$ in comparison to Yang's filter, and only a few pixels are collected by the kernel. Therefore, it suggests that the improved filter better preserves corners and edges in the input image.

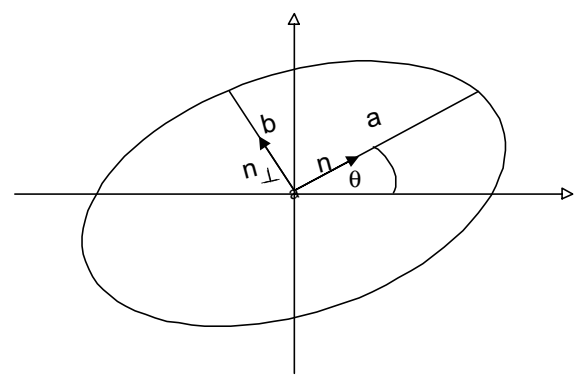

Fig. 1. Controlling the shape and direction of an elliptic kernel through principal axes a, b and direction θ, which are controlled through image local anisotropic features.

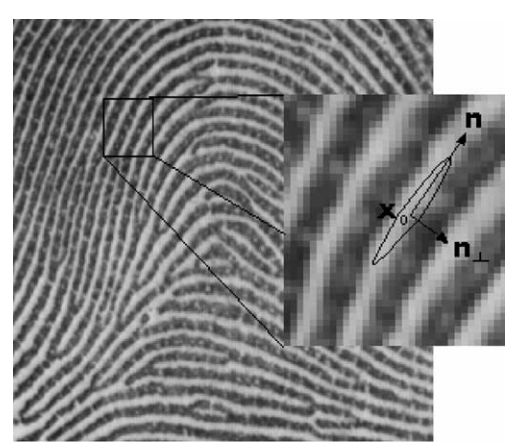

Fig. 2. Controlling the shape and direction of the kernel in the space domain.

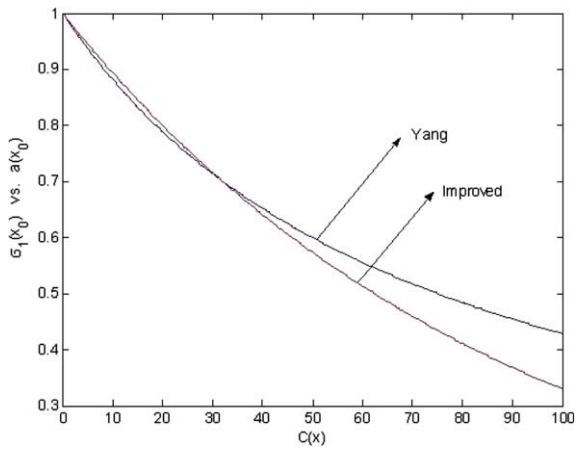

Fig. 3. Controlling the main axis of the filter kernel for Yang's and the improved filter.

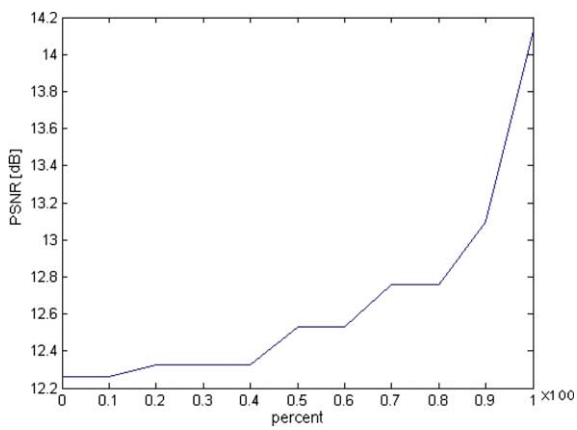

Fig. 4. PSNR of reconstructed Bird image using different β.

Correct estimation of oriented patterns direction is of high importance for effective performance of directional filters such as structure-adaptive anisotropic filter. The technique, which was proposed by Yang et al. (1996), is based on the fact that the power spectrum of an oriented pattern lies along a line through the origin in the Fourier domain, and the direction of the line is perpendicular to the dominant spatial orientation of the pattern. This technique fails to produce correct estimates for noisy images. Wang and Lee (1999) makes use of the directional micro-pattern histograms of fingerprint image for local oriented pattern direction calculation. A directional micro-pattern is defined as 2×2 binary image, in which the white and black pixels construct an oriented pattern. The proposed filter adopts the method, proposed by Donahue and Rokhlin (1993), which uses a gradient-type operator and least-squares minimization to control the noise. The estimation of the oriented pattern direction is extracted from each 2×2 pixels neighborhood, which is then averaged over a local window of 5×5 pixel size.

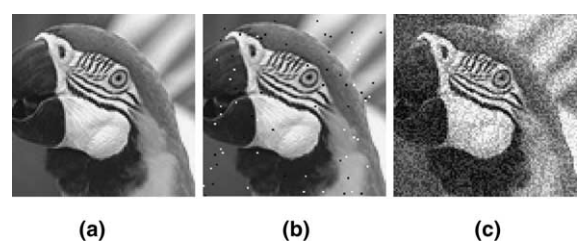

Fig. 7. (a) Original Bird image, contaminated with (b) 'Salt and Pepper' noise with SNR = 21 dB, and (c) Gaussian noise with SNR = 20 dB.

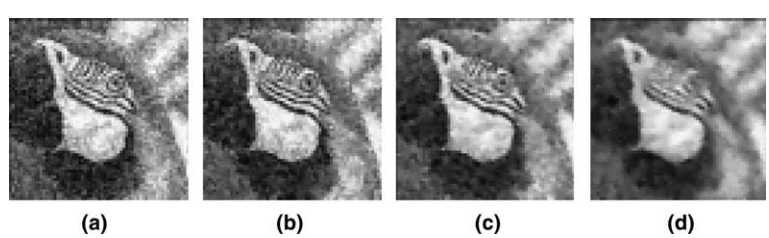

Fig. 5. Result of applying the improved filter with different β: (a) 50%, (b) 75%, (c) 90% and (d) 100% of the maximal value of $c(x)$.

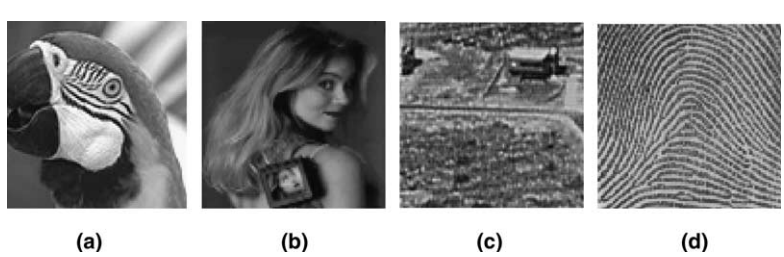

Fig. 6. Test images: (a) Bird, (b) Girl (c) Area and (d) Fingerprint image.

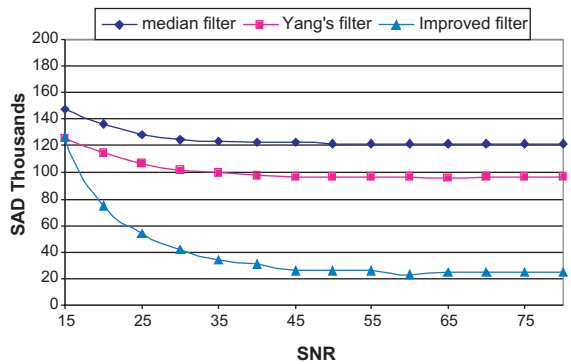

Fig. 8. Performance results of applying a conventional median filter, Yang's structure-adaptive anisotropic and the improved structure-adaptive anisotropic filters on the Bird image contaminated by a Gaussian noise at different SNR levels.

We propose to use a non-linear filter kernel function rather than a linear one that produces less blurring during image filtering. We have implemented the non-linear function in the form of a median filter that is applied within neighborhoods of pixels bounded by an elliptical kernel.

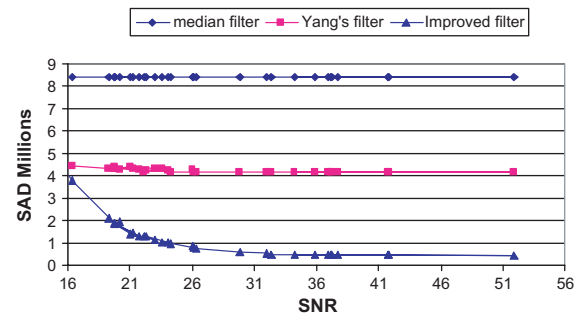

Fig. 9. Performance results of applying a conventional median filter, Yang's structure-adaptive anisotropic and the improved structure-adaptive anisotropic filters on the Bird image contaminated by a 'Salt and Pepper' noise at different SNR levels.

The obtained is the improved structure-adaptive anisotropic filter, which is expressed mathematically as follows:

$$k(x_0, x) = \text{median}\{f(x),\ x \in P(x - x_0)\} \tag{9}$$

where $P(x - x_0)$ is the elliptical kernel centered at pixel x_0, oriented at angle θ (defined by mutually normal unit vectors n and $n_\perp$) and is defined as follows:

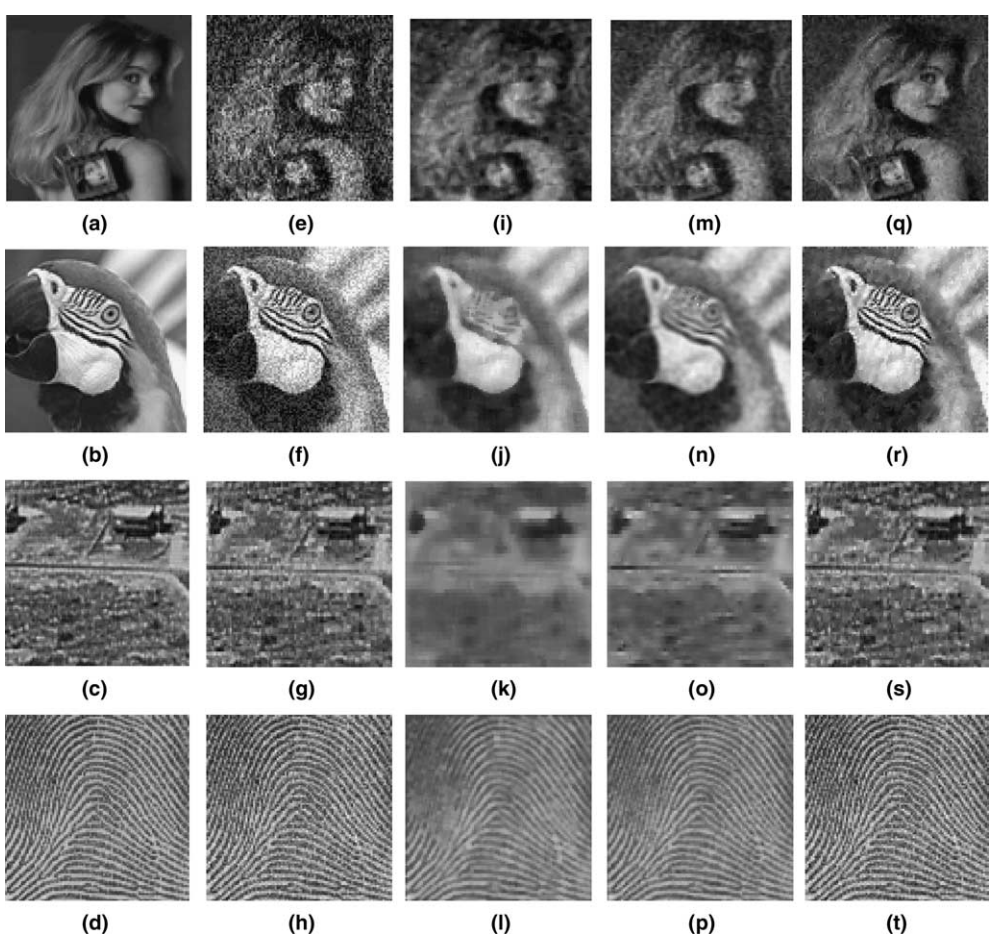

Fig. 10. Applying the different filters on images contaminated by Gaussian noise: Original images (a)–(d), with Gaussian noise (SNR: 10 dB) (e) and (SNR: 20 dB) (f)–(h), images reconstructed by the median filter (i)–(l), by the structure-adaptive anisotropic filter (m)–(p) and by the proposed improved structure-adaptive anisotropic filter (q)–(t).

$$\frac{((x_1 - x_{01}) \cdot n)^2}{a(x_0)^2} + \frac{((x_2 - x_{02}) \cdot n_\perp)^2}{b(x_0)^2} \leqslant 1 \tag{10}$$

Substituting (7) and (8) into (10) and incorporating the obtained equation into (9) results in a final expression of improved structure-adaptive anisotropic filter:

$$k(x_0, x) = \text{median}\left\{ f(x),\ x \in \left(\frac{((x_1 - x_{01}) \cdot n)^2}{\left(r \cdot \exp \cdot \left\{ \frac{-c(x_0)}{\beta} \right\} \right)^2} + \frac{((x_2 - x_{02}) \cdot n_\perp)^2}{\left(r \cdot \exp \cdot \left\{ \frac{-c(x_0)}{\beta} \right\} \cdot (1 - g(x_0) + \varepsilon) \right)^2} \leqslant 1 \right) \right\} \tag{11}$$

Normalization factor β used in the structure-adaptive anisotropic filter (Yang et al., 1996) is set to 75 percent of the maximal value of $c(x)$. Appropriate choice of β involves a trade off between effective smoothing of areas with no corners (higher β) and preserving most of the corners in the image (setting lower β).

Fig. 4 shows the PSNR of a reconstructed image using different β values.

Fig. 4 suggests using higher values of β to obtain higher PSNR values for the reconstructed image. However, the reconstructed image shown in Fig. 5 is over-smoothed while using high value of β. Therefore, in order to preserve most of the corners and to effectively smooth areas without corners, we suggest setting β to 90 percent of the maximal value of $c(x)$. This compromise is empirically achieved by testing different values for β.

4. Experimental results

In this section the performance of the improved structure-adaptive anisotropic filter is compared to the structure-adaptive anisotropic filter (Yang et al., 1996) and to the conventional median filter. The conventional median filter is carried out by numerical sorting of all pixel values in a surrounding neighborhood of 3×3 pixels, and then replacing the pixel being considered with the middle pixel value.

The maximal support radius used for the structure-adaptive anisotropic filter is 3 pixels as suggested by Yang et al. (1996), while the maximal support radius for the improved structure-adaptive filter was empirically set to 2 pixels. The comparison is carried out on different kinds of images, which are commonly used for testing noise filtering algorithms (Fig. 6). The resolution of the test images is 79×79 dots per inch. The original image size is 182×144 pels for the Bird image, 95×95 pels for the Girl and Area images, and 256×256 pels for the Fingerprint image.

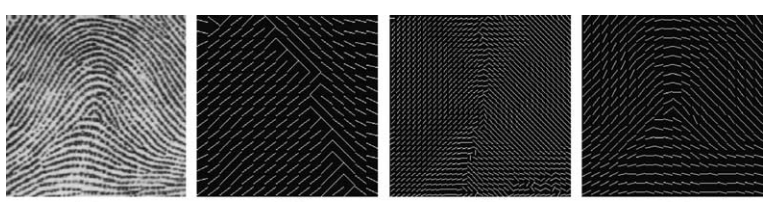

Fig. 11. Results of oriented patterns direction estimation using three different techniques: (a) a Fingerprint image and local ridge orientation estimated by (b) Yang's, (c) Wang's and (d) Donahue's methods.

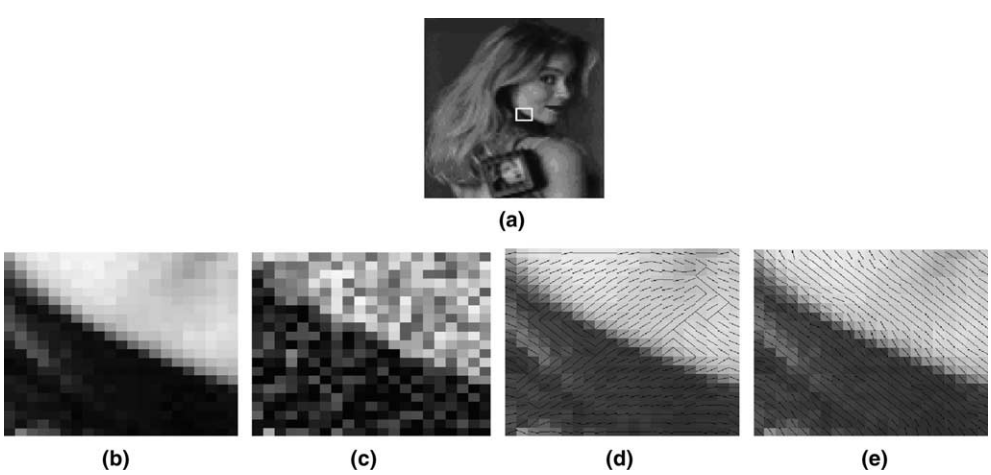

Fig. 12. Results of orientation estimation applied to image contaminated by Gaussian noise. Original image (a) in which the marked white rectangle is zoomed at (b) and contaminated by Gaussian noise with SNR = 12 dB. (c) Results of orientation estimation using both Yang's (d) and Donahue's (e) methods.

The filters were tested on images contaminated by Gaussian noise and 'Salt and Pepper' noise for different SNR (Signal to Noise) levels. Fig. 7 demonstrates the Bird image contaminated with 'Salt and Pepper' noise and Gaussian noise.

The performance comparison is based on the SAD (Sum of Absolute Differences) criteria, which computes the sum of absolute differences between the original image and the reconstructed one. Fig. 8 shows the filters performance results for Gaussian noise for different levels of SNR ranging from 15 dB to 80 dB. Fig. 9 shows the performance results for 'Salt and Pepper' noise for different levels of SNR ranging from 16 dB to 52 dB. Yang's structure-adaptive anisotropic filter performs better than the median filter. The improved structure-adaptive anisotropic filter outperforms both median and Yang's structure-adaptive anisotropic filter over the whole range of SNR levels. Fig. 10 shows some examples of applying the different filters on images contaminated by Gaussian noise. It can be seen that the Yang's structure-adaptive anisotropic filter reconstructs better than the median filter, and the improved structure-adaptive anisotropic filter outperforms both filters in reconstructing from noisy image.

Effective performance of directional filters highly depends on correct direction estimation of oriented patterns. Several techniques for direction estimation of oriented patterns are described in Section 3. We present here, a performance comparison of three techniques by applying them to fingerprint image, in which orientation of ridges is clearly observed. A local window of 5×5 pixels size is used to estimate the oriented patterns for all the methods. In Fig. 11 it can be seen that the Donahue's and Rokhlin method (1993) performs better than the Wang and Lee's (1999) method, and both of them are much better than the method proposed by Yang et al. (1996). Although exact estimation of orientation is a difficult issue in a noisy environment, we conclude that Donahue's method produces much more accurate directions of oriented patterns even for some non-texture images as demonstrated in Fig. 12.

5. Summary and conclusions

We propose to improve the Yang's structure-adaptive anisotropic filter in the space domain. We suggest changing the filter's kernel from a circle to an ellipse with the form, size and direction depending on image local anisotropic features. Instead Yang's derivatives-based estimation of oriented pattern direction we have used a more robust to noise Donahue's method that uses a gradient type local operator and least-squares minimization to control noise. We propose to use a non-linear filter kernel function rather than a linear that produces less blurring during image filtering. We have implemented a non-linear function in the form of the median filter. Comparison results applied to images contaminated by Gaussian noise and 'Salt and Pepper' noise show superior performance of the improved structure-adaptive anisotropic filter over both Yang's structure-adaptive anisotropic filter and the conventional median filter. The improved filter shows enhancement in the filter performance over the whole range of SNR levels. But especially it's effective at high SNR levels, although even at low SNR levels it achieves better visual quality results for reconstructed images.

Effective performance of directional filters is highly depending on correct direction estimation of oriented patterns. Instead of using the technique for oriented pattern direction estimation, proposed in Yang's paper, we suggest using Donahue's technique, which is more accurate and more robust to noise.

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